Venkata Lakshmi Parimala Pasupuleti

Forward Deployed Engineer · Applied AI · Real Estate Domain · LangGraph · RAG · Document Understanding · Python · Claude Code

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Applied AI engineer with **8 years in real estate brokerage and property management SaaS** (Entrata, 100K+ MAU) — the exact operational domain HomeSmart runs. At Intuit, now builds **production AI features** using Anthropic Claude with Python/FastAPI + **Claude Code daily** for scaffolding, refactoring, and debugging. Leasel is a production **8-agent LangGraph platform** for real estate document compliance: **OCR ingestion, section-aware chunking, hybrid RAG, document understanding**, and cited answers — achieving **93.6% precision across 1,240 lease documents**. AgentProbe is an open-source LLM evaluation toolkit routing prompts across 5 providers with CI golden-dataset regression. At Entrata, served as **primary technical contact for external HUD and Tax Credit data integrations**: scoped requirements with partners, built data pipelines, resolved production incidents, and drove go-lives end-to-end — the embedded problem-solver role HomeSmart needs. Stack: **Python, LangGraph, LangChain, FastAPI, RAG, AWS, Docker, GitHub Actions**. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Applied ML: Language Models (Anthropic Claude, OpenAI, Groq, Ollama), LangGraph, LangChain, RAG Pipelines, Document Understanding, PDF Parsing & OCR, Hybrid Search (TF-IDF + Semantic), Embeddings, Vector Search, ChromaDB, Pinecone, AI Agents, HITL, LLM-as-Judge Evaluation, Prompt Engineering, Claude Code

Engineering: Python (9+ yrs), FastAPI, Flask, Node.js, React, TypeScript, Java, Spring Boot, REST APIs, Async Processing (RabbitMQ), SQL Optimization, Data Pipelines, ETL, PostgreSQL, MySQL, Snowflake, Redis, MongoDB, SQLite

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, SQS, API Gateway, IAM), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, Prometheus, Grafana, OpenTelemetry, RBAC & Access Controls, HIPAA/PHI Compliance, Linux

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — Applied AI · Production Features · Financial Analytics

- Build **production AI features** using Anthropic Claude + Python/FastAPI: financial data summaries, natural-language queries, guardrails (input validation, sensitive-data handling, graceful fallbacks); use **Claude Code daily** for scaffolding, refactoring, and debugging AI-powered workflows; deploy via GitHub Actions CI/CD to AWS with Prometheus observability.
- Build **LLM-as-judge eval harnesses** (golden datasets + CI regression) to catch hallucination and drift before every release; build end-to-end ML pipelines (scikit-learn, XGBoost) on Snowflake; **translate ambiguous business requirements into working AI solutions** shipped to production users.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Senior Software Engineer — Real Estate Workflows · Data Integrations · Embedded Technical Partner

- Served as **primary technical contact embedded with HUD and Tax Credit federal data source teams**: identified workflow problems, scoped solutions, built integrations, coordinated go-lives, and resolved production incidents end-to-end — the exact FDE engagement model. Built systems for **income verification, lease compliance, and lender-facing certification workflows** across 1.6M+ property records.
- Shipped 20+ features (Python, PHP, Java, React, PostgreSQL) across **120+ Agile sprints**; **built internal debugging tools that reduced customer issue triage time by 40%**; RabbitMQ async processing at 99.5% uptime; **optimized 200+ SQL queries (30% gain)**; AWS S3 migration with Libsodium encryption and RBAC — 100% PHI compliance; reviewed 300+ PRs, mentored junior engineers.

PROJECTS

Leasel — 8-Agent AI Platform for Real Estate Document Understanding

LangGraph · RAG · Document Understanding

Python · LangGraph (8-agent) · FastAPI · OCR · Hybrid RAG (TF-IDF + semantic) · SQLite · Claude Code — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production real estate AI platform: **PDF parsing pipeline** (OCR ingestion, section-aware chunking) + **hybrid RAG retrieval** over 1,240 unstructured lease documents; 8-agent LangGraph workflow (triage → retrieval → reasoning → validation → HITL escalation); answers cited to exact source clauses; guardrails routing regulated decisions to humans; **LLM-as-judge CI eval: 93.6% precision, 87.9% F1**. Built entirely with **Claude Code**.

AgentProbe — Open-Source LLM Evaluation Toolkit

AI Evaluation · Production Evals

Python · FastAPI · LangGraph · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend routing prompts across 5 LLM providers; **CI golden-dataset regression harness**; LLM-as-judge scoring (accuracy, hallucination, retrieval, completeness, compliance); 3 LangGraph agents; **choose the right model for each job** - frontier API vs. open-weight vs. fine-tuned vs. deterministic logic, weighed against cost, latency, and accuracy.

AI on FHIR — NLP-to-API with Fine-Grained Authorization

Applied AI · Security · Full-Stack

Python · Flask · React · TypeScript · Docker · BERT · LLMs · FHIR REST APIs — github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates EN/ES natural-language queries into live API calls via LLMs + BERT; **HIPAA-style RBAC with fine-grained authorization** (protecting sensitive data in production); Docker containerized; cut manual lookup time ~60%. Demonstrates **security judgment** and production-grade access controls.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · ML pipelines, LLM systems, agentic AI, data governance, unstructured data

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017